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MASTER´S THESIS

Benchmarking Ecological Simulation Models: A Case Study of an Anthropogenic Disturbance Generation SpaDES module

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Abbreviations and Symbols

Abbreviation	Description
AD	Allocation Disagreement
CI / CD	Continuous Integration / Continuous Deployment
CRS	Coordinate Reference System
CSV	Comma-Separated Values
F1	F-score (harmonic mean of precision and recall)
IQR	Interquartile range
LULC	Land Use and Land Cover
NWT	Northwest Territories (Canada)
OAT	One-Factor-at-a-Time (Sensitivity Analysis)
ODD	Overview, Design Concepts, and Details (Model Description Protocol)
POM	Pattern-Oriented Modelling
QD	Quantity Disagreement
QC	Quality Control
RQ	Research Question
SA	Sensitivity Analysis
SpaDES	Spatial Discrete Event Simulation (R package)
TRACE	TRAnsparent and Comprehensive Ecological Modelling (Documentation standard)

Symbol	Description
ρ	Spearman's Rank Correlation Coefficient
nEE	Number of elementary effects
μ^*	Mu-star; mean of absolute elementary effects

Abstract

Rigorous benchmarking and transparent verification are essential to establishing the credibility of ecological simulation models, especially when they are used to inform conservation and land-use decision-making. Yet, many ecological modelling workflows still operate as opaque "black boxes", where undocumented assumptions and untested code pathways make results difficult to reproduce, interrogate, or trust. I present a practical benchmarking workflow—grounded in the TRACE documentation standard and the 'evaludation' framework (a portmanteau of evaluation and validation; Augusiak et al., 2014)—and demonstrate it using the SpaDES (Chubaty & McIntire, 2024) *anthropogenicDisturbance* set of modules (Micheletti, 2025) for the Northwest Territories, Canada.

First, I operationalized the TRACE documentation strategy through ODD-based conceptual documentation, requirements that translate ecological expectations into testable acceptance criteria, and continuous-integration unit tests coupled to a traceability matrix. This makes the model's assumptions, inputs, and computational behaviour explicit and auditable. Second, I used sensitivity screening to identify which parameters and stochastic components most strongly control disturbance outcomes, and then focused uncertainty analysis on those leverage points.

Third, I validated simulated disturbance patterns against standardized observation datasets using map-comparison diagnostics that separate quantity and allocation disagreement and summarize performance across multiple scales. The results show that pixel-perfect prediction of rare disturbance events is an unrealistic benchmark for stochastic landscape generators; instead, credibility is established by capturing plausible rates and emergent spatial structure while clearly identifying systematic mismatches (e.g., disturbance-portfolio composition) that matter for downstream ecological inference.

1 Introduction

Ecological simulation models are increasingly used to explore scenarios, extrapolate beyond observed conditions, and support environmental decision-making (Schmolke et al., 2010). However, the credibility of model-based inference is often undermined by limited transparency: data provenance is unclear, assumptions are implicit, and software implementations are difficult to audit (Grimm et al., 2014; Schmolke et al., 2010). In open natural systems, models cannot be proven “true”; instead, confidence must be built through structured, cumulative evidence of fitness for purpose (Augusiak et al., 2014; Oreskes et al., 1994). Yet the practical skills required to make models transparent—version-controlled code, automated tests, reproducible data pipelines, and traceable documentation—are rarely part of standard ecological training. As a result, ‘good modelling practice’ often remains aspirational, because adoption is constrained less by awareness than by time, incentives, and limited computational training within academic ecology (Micheletti et al., 2024).

This work treats benchmarking as the central problem and contribution. I adopt the 'evaludation' framework—a portmanteau of evaluation and validation—to describe model quality assessment as an iterative workflow spanning data evaluation, conceptual evaluation, implementation verification, model analysis, and output corroboration (Augusiak et al., 2014). In parallel, I use the TRACE documentation logic to make each claim traceable to concrete artifacts (e.g., tests, datasets, and quantitative metrics) rather than narrative reassurance (Grimm et al., 2014, Micheletti et al., 2024).

A key implication for spatial scenario generators is that pixel-perfect prediction is an inappropriate standard. For stochastic landscape generators, fine-grained map overlap is dominated by chance alignment and by persistence in sparse-change landscapes (Pontius et al., 2008; Pontius & Millones, 2011). Benchmarking must therefore emphasize process fidelity—rates, sectoral portfolios, and spatial structures—using multiple, scale-aware criteria that align with downstream ecological questions (Grimm et al., 2005; Grimm & Railsback, 2012; Levin, 1992).

I demonstrate this benchmarking template using the SpaDES *anthropogenicDisturbance* set of modules (Micheletti, 2025), which generates annual anthropogenic disturbance layers (e.g., forestry, mining, settlements, and seismic lines among others) for the Northwest Territories, Canada. The module is used as an input to broader SpaDES-based forecasting studies where disturbance scenarios propagate into habitat and population assessments (e.g., for boreal landbirds and woodland caribou). The case study is therefore intentionally practical: it exposes the common challenges of integrating heterogeneous spatial datasets, stochastic allocation rules, and configuration options within a reproducible simulation workflow.

The remainder of the thesis is organized as a 'how-to' guide for ecologists. Chapter 2 translates benchmarking frameworks into an implementable sequence: document the conceptual model (ODD) (Grimm et al., 2010, 2020), define requirements and traceability, verify implementation with scripted tests, analyze sensitivity and uncertainty, and corroborate outputs with quantitative, interpretable map-comparison benchmarks. Chapter 3 reports the resulting evidence for the *anthropogenicDisturbance* set of modules, and Chapter 4 synthesizes lessons learned and prioritizes next steps based on the observed benchmarking failure modes.

Research questions

Against this background, the thesis is organised around three research questions that mirror the stages of a benchmarking workflow—designing benchmarks, applying them, and learning from them:

RQ1 (benchmarking workflow and verification).

How can the *anthropogenicDisturbance* set of modules be embedded in a transparent, traceable benchmarking framework that links conceptual expectations, model structure, and implementation-level tests? Here I treat verification as demonstrating that the implementation respects its conceptual specification using ideas from TRACE and software-testing practice adapted to SpaDES' modular, discrete-event design (Grimm et al., 2014; SpaDES documentation).

RQ2 (validation and pattern-oriented benchmarking).

To what extent does the benchmarked *anthropogenicDisturbance* set of modules reproduce observed disturbance quantities and spatial allocation patterns in the study area, and what do systematic discrepancies reveal about data limitations or model structure? This question draws on pattern-oriented modelling and the map-comparison literature: I use annualised disturbance rates and the decomposition of disagreement into quantity and allocation components as benchmarks, and interpret them in terms of mis-specified rates or misallocation of disturbance (Grimm & Railsback, 2012; Micheletti et al., 2021; Pontius Jr & Millones, 2011).

RQ3 (uncertainty, sensitivity, and robustness of benchmarks).

How do stochasticity, parametric uncertainty, and scenario choices affect benchmark metrics, and which components of the disturbance generator most strongly control benchmark performance? Building on recent work in uncertainty and sensitivity analysis for environmental models (Pianosi et al., 2016; Saltelli et al., 2007; McIntire et al., 2022), I design ensembles of runs that propagate stochastic and parametric uncertainty through the benchmarking metrics, and I apply global sensitivity screening to attribute variability in benchmark outcomes to key parameters and processes. This allows me to assess not only whether the model meets a given benchmark, but how robust that assessment is to plausible uncertainty.

These questions position the thesis as a practical guide for ecologists who wish to benchmark complex spatial models: starting from data and conceptual expectations, translating them into explicit benchmarks and tests, and then using validation, uncertainty, and sensitivity analyses to arrive at nuanced, fitness-for-purpose judgments rather than binary declarations of success or failure.

2 Methods

2.1 Evaluation framework

To evaluate model quality, I use benchmarking as an organising principle that ties together existing frameworks for ecological model evaluation and documentation. Benchmarking here means deliberately selecting performance criteria, or benchmarks, and using them to compare model behaviour to data and theory in a transparent, repeatable way. This section lays out the conceptual scaffold for that benchmarking workflow, following the evaludation framework (Augusiak et al., 2014), the TRACE documentation standard (Grimm et al., 2014), and pattern-oriented modelling (Grimm & Railsback, 2012), with an explicit focus on ecological modellers who may not have a software-engineering background.

Evaludation proposes that model evaluation should not be a single "validation step" at the end, but an iterative process that spans data evaluation, conceptual model evaluation, implementation verification, model analysis, and model output corroboration (Augusiak et al., 2014). TRACE translates these elements into a structured documentation and evaluation template, emphasising that each decision in model development should be traceable to specific questions, data, and tests (Grimm et al., 2014). Schmolke et al. (2010) similarly argue that models used for environmental decision-making must make their assumptions, uncertainties, and performance explicit if they are to be credible to stakeholders. Together, these works provide the conceptual basis for treating benchmarking not as an ad-hoc comparison, but as a systematic, documented workflow.

I adapt these ideas into a benchmarking workflow with five interlinked components:

1. Data evaluation and selection.

Available datasets (e.g., observed disturbance baselines) are assessed for their suitability as benchmarks: which variables they provide, at what spatial and temporal resolution, and with which uncertainties. This corresponds to evaludation's data evaluation step and forms the foundation for later quantity and spatial-allocation benchmarks (Augusiak et al., 2014; Schmolke et al., 2010).

2. Conceptual model evaluation and ODD-based description.

The intended structure and behaviour of the *anthropogenicDisturbance* set of modules are articulated using an ODD-conformant description (purpose, entities, state variables, processes, and design concepts). This clarifies what the model is supposed to reproduce and which patterns are expected, and therefore which benchmarks are meaningful (Grimm & Railsback, 2012; Grimm et al., 2014).

3. Implementation verification.

Given the conceptual description, I specify and test technical requirements for the SpaDES implementation: event scheduling, correct use of input layers, and reproducibility under fixed random seeds, among others. Here, verification is understood as demonstrating that the implementation is consistent with its conceptual specification, not that the model is "true" in any absolute sense (Grimm et al., 2014; Oreskes et al., 1994).

4. Model uncertainty and sensitivity analysis.

Uncertainty and sensitivity analyses are used to examine how stochasticity, parameter uncertainty, and scenario choices propagate through the benchmarking metrics. This component links model evaluation with global sensitivity analysis and uncertainty propagation frameworks (Saltelli et al., 2007; Pianosi et al., 2016), and supports more nuanced judgments about how robust benchmark performance is across plausible parameter and scenario ranges.

5. Model output verification and corroboration.

Finally, model outputs are compared against the selected benchmarks, focusing first on disturbance quantities and rates, then on spatial allocation patterns. Map-comparison metrics, including the decomposition of disagreement into quantity and allocation components, serve as core benchmarks for assessing whether the model reproduces observed patterns relevant to its purpose (Pontius Jr & Millones, 2011; Grimm & Railsback, 2012). Following Oreskes et al. (1994), I interpret these comparisons in terms of corroboration and fitness for purpose, rather than as strict validation.

These components map directly onto evaludation and TRACE, but I apply them here as a pragmatic benchmarking workflow that can be executed—and re-executed—through scripts, automated tests, and versioned evidence artifacts. The remainder of this chapter follows that workflow: Section 2.2 documents the disturbance generator's conceptual model using ODD; Section 2.3 describes the study area, benchmark disturbance baselines, and supporting spatial layers; Section 2.4 operationalises implementation verification through reproducible system tests, integration tests and targeted checks of core logic; Section 2.5 embeds benchmarking in uncertainty and sensitivity analysis to assess robustness; Section 2.6 defines and applies quantity and spatial-allocation benchmarks across calibration and hold-out periods; and Section 2.7 consolidates these elements into a traceable set of requirements and evidence links. Taken together, this sequence is intended as a transferable, step-by-step template that other ecologists can adapt to benchmark spatial models in similarly code-intensive workflows.

2.2 Model description

Benchmarking requires an explicit statement of what a model is intended to do, which mechanisms it implements, and which observable patterns it should reproduce. For complex, modular simulation models, these expectations are often distributed across source code, configuration defaults, and informal documentation. To make the *anthropogenicDisturbance* set of modules benchmarkable and traceable, I therefore compiled a structured model description using the ODD protocol (Overview, Design concepts, Details), which was originally developed as a standard for describing agent-based and simulation models and has since been updated to include other model types (Grimm et al., 2006, 2010, 2020). This ODD-based description provides the conceptual anchor for subsequent requirements (Section 2.7), implementation verification (Section 2.4), and model output benchmarks (Section 2.6), consistent with TRACE's emphasis on traceability from purpose to tests (Grimm et al., 2014) and evaludation's emphasis on transparent documentation as a prerequisite for evaluation (Augusiak et al., 2014).

2.2.1 Procedure for constructing the ODD description

The ODD description was constructed by triangulating three sources of information: existing module documentation and vignette-style descriptions where available, the executable workflow as expressed in SpaDES configuration and run scripts, and direct inspection of the module code to resolve ambiguities and to capture implicit assumptions (e.g., default parameter values, sequencing of steps, and the treatment of stochasticity). This approach follows TRACE's motivation that model descriptions should be grounded in what the model actually does and should be detailed enough to support independent reproduction and targeted testing (Grimm et al., 2014).

Because the case study workflow is implemented as a set of SpaDES modules rather than a single monolithic program, the ODD "model boundary" is defined at the level of the disturbance simulation workflow: inputs prepared by data-preparation modules are treated as the initial conditions and driving layers for the disturbance generator; the generator's annual outputs (disturbed-area layers by class) are treated as the primary model outputs for benchmarking. Where necessary, the ODD description makes explicit which elements are implemented inside the generator versus provided externally via preprocessed inputs, since this distinction is central for later attribution of benchmark discrepancies to data, assumptions, or generator mechanics (Augusiak et al., 2014; Grimm et al., 2014).

2.2.2 Applying the ODD structure to a modular SpaDES workflow

Following Grimm et al. (2006, 2010, 2020), the ODD description is organised into:

1. Overview (purpose, state variables, and process overview/scheduling).

The "Purpose and patterns" component is used not only to state intended use, but also to name the benchmark-relevant patterns that the workflow is expected to reproduce (e.g., disturbance quantities over time and spatial allocation patterns). The "Process overview and scheduling" component is adapted to SpaDES' discrete-event structure by describing the annual sequence of disturbance generation and post-processing as scheduled events, making clear where stochastic decisions are made and where deterministic spatial operations are applied.

2. Design concepts (key modelling ideas that affect benchmarking).

For the disturbance workflow, design concepts such as stochasticity, spatial heterogeneity (via suitability or "potential" area), and constraints (e.g., avoidance masks) are explicitly described because they determine what kinds of benchmarks are appropriate and how benchmark results must be interpreted. For example, where stochasticity is intrinsic, benchmarking needs to be based on ensembles rather than single-run point comparisons and uncertainty intervals must be computet (Pianosi et al., 2016; Saltelli et al., 2007).

3. **Details** (initialisation, inputs, and submodels).

Initialisation documents how baseline disturbance states and potential/suitability layers are constructed and used at the start of a simulation. Inputs specify the provenance and roles of spatial layers (e.g., disturbance baselines, masks, suitability), linking directly to the data-evaluation step (Section 2.3). Submodels describe the disturbance-generation mechanisms at the level needed to support later verification tests (e.g., conservation constraints, stopping conditions, and reproducibility under fixed seeds), which aligns with the verification–validation distinction emphasised in the modelling literature (Oreskes et al., 1994; Grimm et al., 2014).

The complete ODD description of the SpaDES *anthropogenicDisturbance* Simulation is provided in Appendix A. The purpose of summarising the method here is to make the logic of the benchmarking workflow explicit: ODD is treated as the interface between conceptual expectations (what the model should do) and computational checks (what can be tested and benchmarked), enabling later sections to trace each benchmark metric and verification test back to a clearly documented model claim (Augusiak et al., 2014; Grimm et al., 2014).

For ecologists wishing to benchmark similar workflows, the key transferable step is not the specific disturbance logic but the documentation procedure: define the model boundary and intended patterns, document scheduling and stochastic decision points, and describe mechanisms at sufficient resolution that they can be turned into explicit requirements and automated tests (Grimm et al., 2020; Grimm et al., 2014).

2.3 Data & study area

Benchmarking a spatial disturbance model requires more than "input data": it requires benchmark datasets that can legitimately serve as reference points for evaluating model outputs, plus supporting layers that constrain and contextualize where disturbances can plausibly occur. Consistent with evaludation's emphasis on data evaluation as a first-class evaluation step (Augusiak et al., 2014) and with "fitness-for-purpose" perspectives in predictive ecology (McIntire et al., 2022), this section documents the study area used for the case study, the disturbance baselines used as benchmarking targets, and the preprocessing and harmonization steps that turn heterogeneous geodata into reproducible model inputs and benchmark datasets.

2.3.1 Study area and modelling extent

The case study is implemented for a boreal study area in the Northwest Territories (NWT), Canada. Besides the targeted development of the module series for this region, NWT provides a suitable context for benchmarking an anthropogenic disturbance generator because it contains extensive boreal landscapes with heterogeneous anthropogenic disturbance footprints (linear features such as seismic lines and roads; polygonal features such as cutblocks and settlements), and because SpaDES-based regional landscape simulations have been developed for this jurisdiction, demonstrating the feasibility of integrating multiple spatial processes and datasets at large spatial extents (Micheletti et al., 2021). The modelling extent is treated consistently across all benchmark periods to ensure that comparisons between simulation outputs and observed disturbances reflect model behaviour rather than changing spatial support.

While the module is designed for running across the full NWT extent, I use the full extent primarily for corroboration and “end-to-end” plausibility checks. Verification and sensitivity/uncertainty runs were conducted on a smaller southwest NWT subset to reduce runtime while retaining realistic spatial heterogeneity and real-world data constraints.

In practice, the key benchmarking requirement for the study area definition is consistency of spatial support: the extent and grid onto which all inputs and outputs are mapped must be stable across baseline years, simulation years, and hold-out years, because map-comparison metrics (including quantity vs. allocation decomposition) are sensitive to resolution, projection, and mask definitions (Pontius Jr & Millones, 2011).

2.3.2 Benchmark disturbance baselines (observations used as targets)

To benchmark the disturbance generator, I require observed disturbance layers that include both linear and polygonal anthropogenic features and are available for multiple time points to support both benchmark-period comparisons and temporal hold-outs. Unless otherwise noted, all other experiments in—including routine system tests (and integration runs), sensitivity analysis (SA), and uncertainty analysis (UA)—use the module’s default “production” dataset stack as defined in the DataPrep pipeline and documented in the ODD (Appendix A). In contrast, the core benchmarking targets are derived from Environment and Climate Change Canada (ECCC) anthropogenic disturbance footprint products mapped on a standardized methodology across years (Environment and Climate Change Canada [ECCC], 2013; ECCC, 2019; ECCC, 2024; Government of Canada, 2024).

A critical design choice for benchmarking is that the same disturbance timeline products (2010 → 2015 → 2020) serve both as the empirical basis for deriving disturbance rates and the benchmarking reference for verification and corroboration. This reduces avoidable ambiguity caused by mixing products with differing interpretation rules, class definitions, or mapping scales; it also makes the benchmarking logic transparent because rates and benchmarks originate from the same standardized disturbance-mapping program (Government of Canada, 2024).

BEAD 2008–2010: The 2008–2010 Landsat-interpreted disturbance mapping program provides the earliest baseline used in this workflow. Here it serves as the start state for rate derivation and verification benchmarking of the 2010 → 2015 transition, rather than as an annual-change target (ECCC, 2013; Pasher et al., 2013).

2015 update: The 2015 disturbance layer is treated as the primary anchor year: it is the principal observational benchmark for evaluating whether the disturbance generator reproduces plausible quantities and spatial allocation during the calibration window, and it is also a key input to the empirical rate-derivation logic (ECCC, 2019).

2020 update: A 2020 disturbance layer is used as a temporal hold-out: benchmarking is performed without re-tuning the model to test whether performance generalizes beyond the anchor window (ECCC, 2024). This use of out-of-sample benchmarks aligns with the broader modelling literature's emphasis that corroboration against independent patterns is more defensible than binary "validation" claims (Oreskes et al., 1994), and with the idea that predictive credibility depends on robustness across conditions (McIntire et al., 2022).

Across these baselines, the benchmarking role of each dataset is made explicit: 2010 and 2015 jointly define the empirical transition used for rate derivation and verification-style benchmarking, while 2020 functions as a temporal hold-out for corroboration. Explicit role assignment prevents circularity—i.e., evaluating a model only against the same data used to tune it—while still enabling a coherent "learning-by-example" workflow that ecologists can reproduce (Augusiak et al., 2014; Grimm et al., 2014).

2.3.3 Context, constraint, and supporting layers

In addition to benchmark disturbance targets, the workflow requires layers that define where disturbances can and cannot occur, and layers that provide contextual information for plausibility checks and interpretation. These supporting datasets are not treated as benchmark targets in the map-comparison sense; instead they serve as constraints (e.g., avoidance masks) or as suitability surfaces that guide where simulated disturbances are plausible.

Key examples include:

- Avoidance and constraint masks. Water/wetland masks and elevation constraints are used to restrict unrealistic disturbance placement, consistent with the disturbance generator's design as a suitability- and constraint-based process model. Elevation constraints can be supported using global DEM products such as GMTED2010 (Danielson & Gesch, 2011).
- Suitability / "potential" areas. Beyond exclusion masks, the model also uses "potential" layers to represent where industrial activities are more likely to occur (e.g., areas of higher expected resource-development potential). These layers do not define truth for benchmarking, but they are important for interpreting model behaviour: if allocation errors cluster in low-potential areas, this may indicate missing constraints or mis-weighted suitability rules in the generator.

This separation between "benchmark targets" and "context/constraints/suitability" is deliberate: benchmarking requires clear statements about which datasets are used to judge model performance versus which merely inform generation and interpretation (Augusiak et al., 2014; Grimm et al., 2014).

2.3.4 Harmonization, preprocessing, and provenance

All datasets are harmonized to a consistent spatial support and representation prior to simulation and benchmarking. In this project, these steps are implemented as scripted SpaDES "data prep" modules (one general and one NWT-specific), so that benchmark datasets and model inputs can be regenerated when underlying data versions change. This design choice supports benchmarking as an iterative practice: if new baseline data are released or a preprocessing rule needs revision, the full pipeline can be re-run deterministically rather than reassembled manually (Grimm et al., 2014; McIntire et al., 2022).

Core preprocessing operations include:

1. **Spatial harmonization:** reprojection to a common coordinate reference system; cropping to the modelling extent; and rasterization (where required) onto a common grid used consistently across baseline years and simulations.

2. **Semantic harmonization:** aligning disturbance-class semantics across products and years (e.g., ensuring consistent interpretation of linear features versus buffered footprints).

3. **Conflict resolution and de-duplication:** applying explicit precedence rules when datasets overlap spatially to avoid double-counting disturbance.

4. **Provenance tracking:** recording dataset versions and preprocessing decisions so that benchmark results can be interpreted in light of known uncertainties and data lineage (Grimm et al., 2014; Oreskes et al., 1994).

These steps are necessary because map-comparison benchmarks (e.g., quantity disagreement and allocation disagreement) assume that the compared maps share a common spatial domain and consistent class semantics (Pontius Jr & Millones, 2011). Moreover, ECCC's disturbance products explicitly document methodological caveats relevant to interpretation—for example, overlaps among disturbance classes and potential underestimation of some linear features due to mapping hierarchy and overlap-handling—so benchmarking results must be read with those data limitations in mind (Government of Canada, 2024).

For ecologists benchmarking similar spatial models, the transferable data lesson is to assign explicit roles to datasets (rate derivation vs. benchmark target vs. constraints/suitability), harmonize spatial support and class semantics before computing metrics, and formalize overlap-resolution and provenance rules in code so that benchmark outcomes remain interpretable and reproducible across versions (Augusiak et al., 2014; Grimm et al., 2014).

2.3.5 Data availability, licensing, and credibility considerations

Benchmarking is only as credible as the data used to define "what happened." I prioritize disturbance baselines that are produced by a federal authority using a documented and repeatable mapping methodology and are distributed under an open licence suitable for scientific reuse (ECCC, 2013, 2019, 2024; Government of Canada, 2022).

Supporting layers used for constraints and suitability are drawn from similarly transparent sources where possible (e.g., government open data portals and well-documented global products). At the territorial level, many contextual layers are distributed via the GNWT Open Data Portal under the Open Government Licence – Northwest Territories (Government of Northwest Territories, 2022; Government of Northwest Territories, n.d.).

Where global products are used (e.g., land cover), the workflow prefers resources with explicit citation guidance and permissive reuse terms; for example, ESA WorldCover provides a citable 10 m land-cover dataset with a stated open licence (Zanaga et al., 2021).

2.4 Implementation verification

Implementation verification is a central quality step because the credibility of any subsequent benchmarking against observational data depends first on the integrity of the code. In evaludation, implementation verification is defined as "testing the model's implementation in equations and as a computer program" (Augusiak et al., 2014). In practice, this means systematically checking for programming errors and confirming that the implemented algorithms behave in accordance with the conceptual model description (Section 2.2). This emphasis is motivated by the epistemological constraint that ecological simulation models cannot be proven true through "validation" against reality in any absolute sense (Oreskes et al., 1994). Instead, credibility must be built through transparent documentation, reproducible workflows, and thorough testing of the implementation prior to and alongside output corroboration (Grimm et al., 2014).

Implementation verification in this study combines version control and traceability practices that make changes auditable, automated regression testing at the system level, targeted synthetic verification scenarios that test core logic under controlled conditions, and reproducibility, logging, and diagnostic instrumentation that help detect and explain failures. Collectively, these measures align with TRACE's emphasis on transparency and evidence for thorough testing during model development (Grimm et al., 2014) and with broader guidance for ecological models used in decision contexts, where reliability and clear documentation are prerequisites for credible use (Schmolke et al., 2010).

2.4.1 Version control, provenance, and traceability

All code used in this work in is maintained in a Git repository, providing a complete and auditable history of changes through commits and branches. Version control supports implementation verification in two ways. First, it enables reproducibility of results by allowing any simulation or test outcome to be associated with an exact code revision. Second, it supports traceability by making it possible to link requirements and benchmarking outcomes to the precise implementation state that produced them (Grimm et al., 2014).

To strengthen this linkage, a traceability matrix was developed to map model requirements and claims to specific code modules, test cases, and evidence artifacts (Section 2.7). This approach operationalizes TRACE's goal that model development and evaluation be documented in a way that can be audited step-by-step, including how tests support claims about model behaviour (Grimm et al., 2014).

2.4.2 Automated system test harness and regression testing

The primary implementation verification mechanism is an automated system-level test harness that executes the disturbance workflow end-to-end across a matrix of scenarios. Each scenario specifies a controlled set of inputs and parameter settings, runs the full simulation pipeline, and records pass/fail status and diagnostic outputs. This design includes verification of integrated behaviour, the interaction of multiple functions and spatial operations, rather than isolated function-level correctness, reflecting the emergent nature of spatial simulation workflows where failures often occur in interactions (Augusiak et al., 2014; Grimm et al., 2014).

The test harness functions as a regression test suite: the same scenarios are re-run across code revisions to detect unintended behavioural changes. This is consistent with TRACE guidance that code should be "thoroughly tested for programming errors" and that testing should be documented as evidence of implementation quality (Grimm et al., 2014). In the context of benchmarking, regression tests provide an explicit guardrail against "silent drift" in model behaviour, ensuring that improvements or refactors do not inadvertently change disturbance rates, spatial constraints, or stochastic mechanisms in ways that would confound interpretation of benchmark results.

2.4.3 Synthetic scenario verification of core model logic

System tests are complemented by targeted verification scenarios using synthetic or idealized inputs designed to test specific aspects of model logic under controlled conditions. For example, synthetic disturbance cases are constructed to check whether realised disturbed area aligns with the disturbance rates and stopping rules specified by scenario inputs.

These targeted checks function as “logic verification” tests: they are designed to confirm that key algorithms behave as implied by the model description, even in simplified conditions. This is particularly important for components where stochasticity or spatial constraints can obscure whether the underlying implementation is correct. In evaludation terms, such tests address whether “the implemented model performs as indicated by the model description” (Augusiak et al., 2014), and they strengthen the interpretability of later benchmarking by ensuring that model–data disagreements are not trivially explained by implementation errors.

2.4.4 Reproducibility controls for verification runs

Implementation verification requires that test outcomes be attributable to code behaviour rather than uncontrolled variability. For this reason, verification scenarios support fixed random-number seeds so that stochastic components produce repeatable outcomes when required. Seeds and scenario settings are recorded in run logs and config files to allow exact reproduction of each verification case.

In addition, the software environment is stabilized through dependency management. Package versions are locked using a `renv` snapshot to reduce variability in system libraries and geospatial dependencies across machines. These measures reduce the risk that verification failures (or apparent performance changes) reflect environmental drift rather than changes in code, which is essential for traceable and credible benchmarking (Grimm et al., 2014).

2.4.5 Logging, diagnostics, and fail-fast assertions

To make verification actionable, the workflow implements extensive logging and runtime diagnostics. Each test scenario records timestamps, parameter configurations and seeds, along with error messages when failures occur. A “fail-fast” approach follows recommended practice in model development: embedding diagnostic code that stops execution or outputs a clear warning message when internal model states leave expected ranges improves the ability to detect and localize implementation problems (Grimm et al., 2014). From a benchmarking perspective, comprehensive logging and diagnostic output also provides the evidence trail required for TRACE-style documentation of how the code was tested and how failures were resolved (Grimm et al., 2014).

2.4.6 Code review and independent scrutiny

The literature notes that implementation verification can be strengthened by independent scrutiny such as peer code review or independent re-implementation of critical components (Grimm et al., 2014). In this project, the primary mechanism for independent scrutiny is behavioural cross-checking through system tests and synthetic verification scenarios rather than a formalized peer-review protocol. The absence of a documented, structured peer code review is therefore treated as a limitation of the verification process and a clear opportunity for improvement, especially for critical sections that strongly influence benchmark outcomes (e.g., disturbance placement logic and rate application).

2.4.7 Continuous integration as ongoing implementation verification

Implementation verification is most effective when it is continuous rather than episodic. To operationalize this principle, the verification suite is being integrated into a continuous integration (CI) workflow so that core tests can be executed automatically on code changes. Given computational constraints, the CI design emphasizes lightweight verification cases (unit tests, selected system-test scenarios, and key synthetic checks), while heavier benchmarking suites (large ensembles, extensive uncertainty and sensitivity analysis) are reserved for scheduled or release-based execution (Augusiak et al., 2014; Grimm et al., 2014). This division supports the practical goal of catching regressions early while maintaining feasible compute budgets, and it reinforces the thesis' overarching framing of benchmarking as an iterative workflow rather than a one-time evaluation.

Overall, this implementation verification strategy provides a structured foundation for the subsequent benchmarking and corroboration steps (Section 2.6) by maximizing confidence that model outputs are generated by the intended algorithms, under controlled and reproducible conditions, with documented evidence of testing and error detection (Augusiak et al., 2014; Grimm et al., 2014; Oreskes et al., 1994).

2.5 Uncertainty and sensitivity analysis

Benchmark metrics are not fixed properties of a model in the abstract; they are outputs of a stochastic and parameterised simulation system confronted with imperfect observational benchmarks. For this reason, benchmarking results can be misleading if they are reported as single numbers from single runs, without addressing how much those numbers vary under plausible uncertainty. Uncertainty analysis (UA) and sensitivity analysis (SA) are therefore treated as integral components of benchmarking: UA quantifies how benchmark outcomes vary due to stochasticity and uncertain inputs, while SA identifies which assumptions and parameters most strongly control benchmark performance (Pianosi et al., 2016; Saltelli et al., 2007). This aligns with recent calls to treat predictive ecology as a discipline of robust inference rather than point prediction (McIntire et al., 2022).

2.5.1 Sources of uncertainty relevant to benchmarking

Following common classifications in environmental modelling, uncertainty is organised into three categories that directly influence benchmarking metrics (Pianosi et al., 2016; Saltelli et al., 2007):

1. **Stochastic uncertainty (intrinsic randomness).** The disturbance generator includes stochastic components (e.g., random selection or placement within suitability-defined candidate areas, stochastic sampling of event locations or sizes). For such processes, any single run represents one realisation; benchmark outcomes (e.g., disagreement metrics) are therefore random variables that must be summarised across replicates (Pianosi et al., 2016).
2. **Parametric uncertainty (imperfectly known parameters).** Imperfectly known rates and size/spacing parameters can shift both quantity and allocation outcomes.
3. **Data and benchmark uncertainty (imperfect observations).** Benchmark layers derived from remote sensing or compiled geodata can include omission/commission errors, temporal mismatch, and class-definition inconsistencies. Consequently, disagreement between simulated and observed

maps reflects a combination of model limitations and benchmark uncertainty. This reinforces the use of corroboration language and fitness-for-purpose judgments rather than claims of truth (Oreskes et al., 1994).

This categorisation is used operationally: it determines what is varied in UA, what is screened in SA, and how benchmark outcomes are reported (e.g., as distributions rather than point estimates).

2.5.2 Sensitivity analysis design

SA addresses a distinct benchmarking question: which parameters or design choices control benchmark outcomes? This matters for two reasons. First, it identifies which assumptions deserve the most scrutiny and documentation (TRACE), because they dominate benchmark performance. Second, it guides efficient model improvement by indicating where parameter refinement or structural changes are most likely to improve benchmark fit (Grimm et al., 2014; Pianosi et al., 2016).

Comprehensive variance-based global sensitivity analysis can be computationally expensive for spatial, long-horizon simulations. For this reason, the SA here prioritises global screening approaches suitable for high-dimensional models, consistent with recommended workflows in environmental modelling where screening is used to identify the small subset of influential inputs before deeper analysis (Saltelli et al., 2007; Pianosi et al., 2016).

A key design choice is that SA is performed on the same outputs used for benchmarking (e.g., class-specific disturbed area and configuration metrics). This ensures that sensitivity results are directly actionable: the “most sensitive inputs” are those that most affect the model’s benchmark performance, not merely intermediate internal states (Pianosi et al., 2016).

Inputs with high influence on benchmark metrics are treated as priority targets for improved parameterisation, clearer documentation of assumptions, and robustness checks in UA. Conversely, inputs with consistently low influence can be deprioritised, reducing unnecessary parameter-tuning effort and supporting more transparent communication of what matters in the model (Saltelli et al., 2007; Grimm et al., 2014).

2.5.3 Uncertainty analysis design

The objective of UA is not only to quantify output variability, but to quantify variability in the benchmark metrics themselves (e.g., disturbed area bias). This supports more defensible benchmarking statements such as: "Observed disturbance lies within the simulated uncertainty envelope" rather than relying on a single run (Pianosi et al., 2016; McIntire et al., 2022).

The UA design uses ensembles of replicate runs under fixed configurations and/or systematically varied stochastic settings. Replicates are generated by varying random seeds while holding other inputs constant, which isolates stochastic uncertainty. Where additional stochastic or discrete modelling options exist, UA is extended by varying those options to evaluate their effect on benchmark metrics, consistent with the idea that benchmarking should test robustness to plausible modelling choices (Augusiak et al., 2014; Pianosi et al., 2016).

UA results are summarised as distributions (e.g., median and uncertainty intervals) for a set of benchmark metrics and benchmark years/periods. The practical benchmarking implication is that comparisons against observations can be framed probabilistically: whether observed quantities fall within the ensemble range, and whether conclusions are stable across replicates (Pianosi et al., 2016).

In evaludation terms, UA contributes to model analysis and strengthens corroboration by ensuring that benchmark conclusions are not artefacts of a single stochastic realisation (Augusiak et al., 2014).

2.6 Verification benchmark metrics and corroboration of model outputs

Benchmarking requires that evaluation criteria be made explicit before results are interpreted. For a spatial anthropogenic disturbance generator, benchmarks must address at least two distinct questions: whether the model produces the right amount of disturbance (quantity), and whether it places disturbance in plausible locations and reproduces spatial patterns relevant to ecological processes (allocation and configuration). This distinction reflects long-standing arguments in landscape ecology that spatial pattern matters for ecological function (Turner, 2010) and is consistent with pattern-oriented modelling, which advocates confronting models with multiple patterns at different scales to reduce the risk of "getting the right answer for the wrong reason" (Grimm & Railsback, 2012).

Model outputs were evaluated using a set of complementary benchmarks that are interpretable for practitioners and traceable to the conceptual model description (Section 2.2) and requirements (Section 2.7). Following the epistemological position that models cannot be proven true, benchmark comparisons are interpreted as corroboration (or lack thereof) relative to the chosen benchmarks and their uncertainties, rather than as definitive validation (Oreskes et al., 1994).

2.6.1 Rationale and selection of benchmarking metrics

The choice of benchmark metrics is guided by three criteria: fitness for purpose, diagnostic value, and practical computability. Fitness for purpose requires that benchmarks align with the intended use of the disturbance generator in broader ecological assessments (e.g., representing disturbance pressure and fragmentation drivers). Diagnostic value requires that benchmark results help identify why a mismatch occurs (e.g., incorrect rates versus misallocation), rather than producing a single opaque accuracy score. Practical computability requires that metrics can be computed reproducibly over large landscapes and long simulations without excessive manual intervention, supporting iterative benchmarking and TRACE-style documentation (Augusiak et al., 2014; Grimm et al., 2014).

Accordingly, the benchmarking suite is structured as a sequence of checks:

1. **Quantity benchmarks (first-order):** Does the model reproduce disturbance totals and rates by class and time window?
2. **Allocation benchmarks (second-order):** Given broadly correct totals, does the model reproduce where disturbance occurs?
3. **Hold-out corroboration (generalisation):** Do these benchmark conclusions hold when evaluated against an independent time period?

This sequence reflects a practical modelling logic: if quantities are wrong, spatial metrics may be dominated by rate mis-specification; conversely, if quantities are approximately correct but allocation disagreement remains high, the model may be structurally plausible but mis-specified in its spatial suitability and constraints. For allocation benchmarking, I adopt the decomposition of map disagreement into quantity disagreement and allocation disagreement as proposed by Pontius Jr and Millones (2011). This decomposition is particularly useful in benchmarking because it separates "how much" from "where," providing a direct diagnostic link to model components (e.g., rates versus suitability).

Finally, consistent with evaludation and TRACE, benchmark results are not treated in isolation. They are interpreted in the context of known sources of uncertainty in both model outputs (stochasticity and parameter uncertainty) and benchmark datasets (classification error, temporal mismatch, incomplete detection), and are complemented by uncertainty and sensitivity analysis in Section 2.5 (Augusiak et al., 2014; Pianosi et al., 2016; Saltelli et al., 2007; Grimm et al., 2014).

2.6.2 Benchmarking disturbance quantities and rates

Quantity benchmarks assess whether the model reproduces the magnitude of anthropogenic disturbance by class (e.g., linear versus polygonal disturbance categories) and by time period. These benchmarks are the most direct test of whether the disturbance generator's rate parameters and spatial constraints produce plausible outcomes at the scale relevant for downstream ecological modelling (Turner, 2010).

For each benchmark year (and/or time window), simulated and observed disturbances are summarised as:

- total disturbed area (or effective footprint) per disturbance class;
- annualised disturbance rates derived from the change over the time window;
- total line length per linear disturbance class.

Where linear disturbances are represented as buffered footprints, quantity benchmarks are computed on the buffered area, because ecological effects (edge, fragmentation, effective habitat loss) are more closely aligned with footprint than with centreline length. The specific buffering assumptions are documented as part of the benchmarking specification, because they can materially affect both quantity totals and map comparisons (Turner, 2010). In these benchmarks buffers around linear disturbances were fixed at 30m.

Quantity benchmarks require consistent spatial support and class semantics between simulated and observed maps (Section 2.3). To prevent circularity, I distinguish between baselines used for initialisation or rate derivation and baselines used as benchmarking targets, and I interpret results accordingly (Augusiak et al., 2014). In particular, when a benchmark layer also informs model parameterisation, quantity comparisons are interpreted as internal consistency checks rather than as independent corroboration; the stronger test is temporal hold-out corroboration (Section 2.6.4).

2.6.3 Benchmarking spatial allocation: quantity and allocation disagreement

Spatial allocation benchmarks evaluate whether the model reproduces *where* disturbance occurs, beyond matching totals. This matters because ecological consequences depend on spatial configuration (e.g., clustering versus dispersion of disturbance, fragmentation) (Turner, 2010).

To separate amount from placement, I use Pontius Jr and Millones' (2011) decomposition of categorical map disagreement into quantity disagreement (QD) and allocation disagreement (AD). Conceptually, QD measures disagreement due to differences in class proportions (how much of each class), while AD measures disagreement due to spatial mismatch given the class totals (where classes occur). This is particularly well-suited for benchmarking because the two components map naturally onto different parts of the disturbance generator: rates and stopping rules primarily influence QD; suitability surfaces, spatial constraints, and the stochastic placement mechanism primarily influence AD.

For each benchmark comparison, simulated and observed maps are harmonised to the same spatial support and classified into comparable categories (Section 2.3). A confusion matrix is constructed between the simulated and observed categories. From this matrix, QD and AD are computed following Pontius Jr and Millones (2011).

In benchmarking practice, QD and AD support an interpretable diagnostic logic (Pontius Jr & Millones, 2011):

- High QD, low AD: the model primarily fails in amount (rates/constraints), but given its totals it places disturbance relatively plausibly. Benchmark implication: prioritise revisiting rate derivation, stopping rules, and potential-area definitions before revising suitability.
- Low QD, high AD: the model gets totals approximately right but places disturbance incorrectly. Benchmark implication: revisit spatial suitability drivers, accessibility assumptions, constraints, and the stochastic placement mechanism.

- **High QD, high AD:** both amount and placement are problematic. Benchmark implication: treat as a structural mismatch or compounded data/model issues; revisit benchmark dataset assumptions as well as core generator logic.
- **Low QD, low AD:** the model reproduces both amount and spatial allocation sufficiently for the benchmark's purpose, subject to uncertainty and hold-out corroboration.

This decomposition therefore serves as a "bridge" between map-comparison statistics and model-development decisions, supporting the evaludation objective of making evaluation results actionable rather than purely descriptive (Augusiak et al., 2014; Grimm et al., 2014).

2.6.4 Corroboration using temporal hold-out benchmarks

A central risk in benchmarking is circularity: if the same data that inform model parameterisation are also used as evaluation targets, apparent "good performance" may reflect tuning rather than generalisable predictive skill. To address this, I apply the same benchmark metrics to a temporal hold-out period that was not used for parameter derivation or calibration decisions. This aligns with the view that model evaluation should focus on corroboration against independent patterns rather than definitive validation (Oreskes et al., 1994).

The hold-out benchmark is treated as a strict "no retuning" test: the benchmarking scripts, class definitions, and metric computations are kept identical to the calibration-period benchmarking, and only the observational reference map changes to the hold-out year. This ensures that differences in benchmark outcomes reflect temporal generalisation rather than changes in evaluation machinery (Grimm et al., 2014).

Hold-out corroboration is interpreted in terms of whether benchmark conclusions are stable:

- If the model performs similarly (within uncertainty) on both calibration-period and hold-out benchmarks, this strengthens confidence that the disturbance generator captures stable mechanisms relevant to the study area.
- If performance deteriorates markedly on the hold-out, this suggests non-stationarity (e.g., changing disturbance drivers), structural mismatch, or benchmark-data inconsistencies between periods, and motivates revisiting

assumptions about rate stationarity, suitability drivers, and data comparability (Schmolke et al., 2010; Oreskes et al., 1994).

The hold-out benchmark therefore functions as the most policy-relevant benchmarking step: it tests whether the disturbance generator can plausibly be used beyond the period that informed its configuration, which is essential when the model is intended for scenario analysis and projection rather than retrospective reconstruction (McIntire et al., 2022; Schmolke et al., 2010).

For ecologists adopting a similar approach, the key transferable point is to treat benchmarking as a staged process: begin with quantity plausibility checks, then evaluate allocation via interpretable decompositions such as QD/AD, and finally test generalisation with an explicit hold-out where the evaluation machinery is held constant (Pontius Jr & Millones, 2011; Augusiak et al., 2014; Grimm et al., 2014).

2.7 Requirements and traceability

Because benchmarking is only persuasive when readers can trace each claim about model quality to concrete, inspectable evidence, I operationalized "requirements and traceability" as a living, auditable bridge between what the disturbance generator is intended to do, how I check that it does so correctly, and which artifacts constitute evidence. This follows the rationale behind TRACE documentation: ecological decision support benefits from transparent reporting of model purpose, testing, and analysis so credibility can be assessed by others rather than asserted by authors (Grimm et al., 2014; Schmolke et al., 2010). It also aligns with "evaludation" framing, where model quality is established through multiple complementary checks (conceptual consistency, implementation verification, and output-based evaluation), each tied to explicit evidence (Augusiak et al., 2014). At the same time, I treat traceability as a practical response to a core limitation of modelling in open natural systems: absolute verification/validation is not attainable, so the goal becomes a structured, updateable record of what has passed, failed, or lacks evidence (Oreskes et al., 1994). Concretely, I maintained a version-controlled requirements table as the source of truth: each requirement is stated as a verifiable claim with an acceptance rule and one or more pointers to evidence and/or summary metrics, and the traceability matrix is generated automatically from that table. Each evidence pointer is resolved by checking presence and interpreting standard artifact types (e.g., unit-test outcomes and coverage summaries; configuration/schema validation outputs; and curated snapshots of computationally expensive UA/SA results), which enables continuous integration to refresh verification status. In practice, this “fast checks in CI + snapshot heavy evidence” pattern was a key lesson learned: it preserves reproducibility and auditability under realistic compute constraints while still keeping long-running benchmarking results linked and visible (McIntire et al., 2022). The matrix also supports accessibility for ecologists by making the workflow explicit and repeatable: derive requirements from the model specification (e.g., ODD-derived behavioural and structural expectations), define acceptance criteria and attach evidence types that can be generated routinely (Grimm et al., 2006, 2010, 2020). Finally, because benchmarking commonly depends on quantitative comparison summaries (including map agreement and change-accounting), the same traceability mechanism can point to compact, interpretable metrics (e.g., allocation vs. quantity disagreement) alongside pass/fail checks, keeping evaluation criteria explicit and comparable across runs (Aldwaik & Pontius, 2012; Pontius & Millones, 2011).

3 Results

3.1 Sensitivity Analysis

Given the computational cost of the disturbance generator (approximately 30–50 minutes wall-time per replicate on a hardware-constrained local workstation, implying on the order of ~3 hours per design point when four replicates are executed sequentially), sensitivity analysis was implemented as a screening step to triage which assumptions warrant deeper scrutiny, rather than as an exhaustive variance decomposition. I therefore used a Morris elementary-effects design as a pragmatic accuracy–performance trade-off, varying four candidate high-leverage factors over defensible bounds: `totalDisturbanceRate` (0.5–3.0), `clusterDistance` (500–2000), `useClusterMethod` {FALSE, TRUE}, and `siteSelectionAsDistributing` (ordered set from none to oilGas; cutblocks; mining; seismicLines).

The design used r = 10 trajectories on a p = 6 level grid with grid_jump = 1, yielding a normalized step size of $\Delta = 0.2$; continuous factors were mapped by linear scaling from [0,1] into their bounds, while discrete factors were mapped by rounding to the nearest level. The executed experiment evaluated 10 trajectories × 6 points × 4 replicates = 240 runs, summarized into 7 disturbance metrics at two benchmark years (2021, 2031) (14 metric–years). Here we use Morris μ^*, the mean absolute elementary effect, interpreted as a factor's overall influence on an output. During analysis, the initially generated Morris steps were found to violate one-at-a-time (OAT) assumptions—some step pairs changed no analysed parameter and others changed multiple parameters—so the final results are reported only for a quality control (QC) salvaged executed design that retains step pairs with exactly one analysed factor change (50 potential step pairs; 33 kept, 13 no-op, 4 confounded). Mechanistically, the no-op steps arose because discrete factors were mapped by rounding, so a planned Δ step did not always change the realized discrete level; in addition, an “excluded” parameter (distanceNewLinesFactor) varied in executed configs despite being omitted from the analysed factor set, contaminating some transitions.

After QC, effective elementary-effect support was uneven across factors (`clusterDistance` nEE = 11; `siteSelectionAsDistributing` nEE = 10; `totalDisturbanceRate` nEE = 10; `useClusterMethod` nEE = 2), meaning conclusions about `useClusterMethod` from Morris alone must be treated as provisional and corroborated. Despite these caveats, a consistent dominance pattern emerged at the scale of the chosen summary metrics: `totalDisturbanceRate` ranked #1 by μ* in 9/14 metric–years overall and dominated polygonal outcomes in 6/8 polygon metric–years (e.g., total polygon area and mean patch area), while `clusterDistance` was the top driver for polygon patch count in 2/8 polygon metric–years, consistent with threshold behaviour in how polygons fragment/aggregate. `siteSelectionAsDistributing` was consistently low influence (ranked 3rd–4th across all metric–years), indicating that, at least for these aggregate benchmarks, site-selection stochasticity is secondary to global rate and clustering controls. Linear outcomes split into two regimes: `useClusterMethod` and `totalDisturbanceRate` each ranked #1 in 3/6 linear metric–years, but because `useClusterMethod` had only two QC-valid elementary effects, I treated it as a candidate structural switch and validated that interpretation with independent analyses: the scenario split (`useClusterMethod` 0=FALSE, 1=TRUE) showed large, systematic differences in seismic line outputs, with median total_linear_length in 2031 shifting from approximately 205,654 km (FALSE) to 13,983 km (TRUE) (and in 2021 from 32,533 km to 12,236 km), alongside similarly large shifts in segment counts and mean segment lengths (Fig. 1). Regression corroboration on executed design summaries attributed most explainable variance in total_linear_length to `useClusterMethod` (partial $R^2 \approx 0.68$ in 2021 and 0.87 in 2031 in a main-effects model), confirming that this setting fundamentally changes how seismic lines are generated and should be treated as a scenario-level modelling choice rather than a minor tuning parameter.

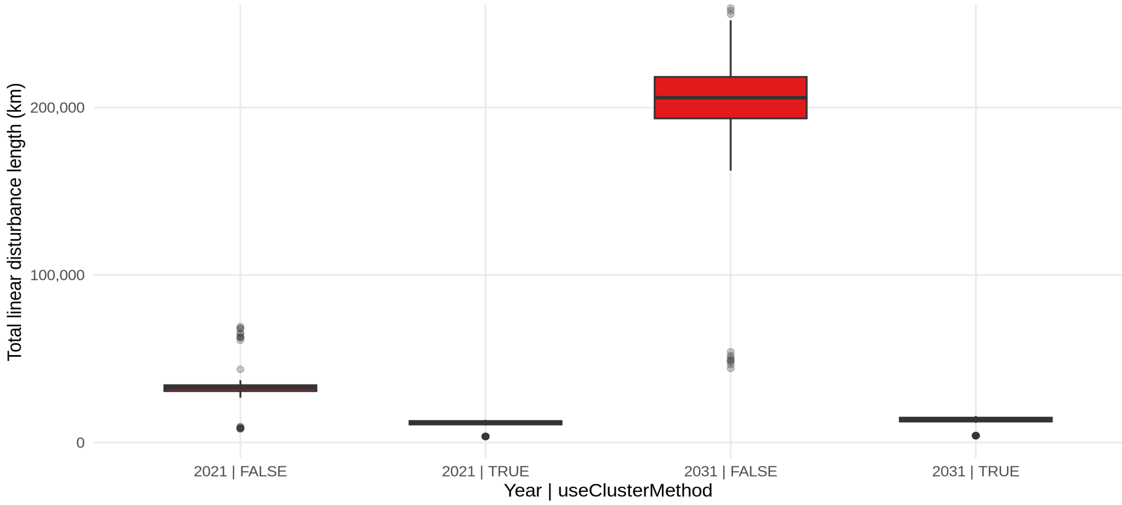

Figure 1: Total simulated linear disturbance length (km) from the Morris screening experiment, stratified by benchmark year (2021, 2031) and the structural switch useClusterMethod. Each boxplot summarizes the distribution across executed screening runs for that group (boxes = median and IQR; whiskers = 1.5×IQR; points = outliers, n = 76–164 runs per group).

Collectively, the Morris screening (with QC salvage and corroboration) served its intended benchmarking role: it identified which uncertainties materially control benchmark-relevant disturbance outputs under a constrained compute budget, and it directly motivates the downstream uncertainty analysis to focus on stochastic variability conditional on a regime, and parametric uncertainty concentrated on the dominant drivers particularly `useClusterMethod` as a regime switch for linear disturbances.

3.2 Uncertainty Analysis

Building on the screening step in Section 3.1.1, the uncertainty analysis shifts the benchmarking question from *"what controls the outputs?"* to *"how much variation should I expect when those controls (and intrinsic stochasticity) are present?"* For stochastic models, benchmarking is most defensible when it reports an ensemble-based expectation rather than a single run: the practical goal is to establish a baseline envelope (what changes across stochastic realisations under a fixed configuration) and then identify when configuration/parameter choices move outcomes beyond that envelope. I implemented this in a simple, reproducible workflow that is transferable to other ecological simulations: run ensembles at fixed inputs and summarise key generator outputs as median and uncertainty interval (p05–p95) for seed ensembles or as between-design ranges for configuration/parametric ensembles. Table 1 consolidates the benchmark-relevant 2031 outcomes.

Table 1: Uncertainty analysis (UA) experiment designs and 2031 outcome distributions across three experiment types (seed-only, configuration, and parametric/design uncertainty). For each output metric, values are reported with units (km^2 or km) and summarized as median [5th–95th percentile] across runs (n as indicated). Outputs that are deterministic under the experiment design are labelled as such.

Experiment / design context	What varies and what is held constant	Key output (2031)	Summary statistic
UA base (seed-only)	Seeds vary; configuration fixed. *(One replicate excluded due to missing 2031 outputs; summary uses complete runs.)*	Total interval new area (km^2)	159.46 [157.22–161.63] *(n=9)*
		Forestry cutblocks total area (km^2)	246.13 [246.13–246.13] *(deterministic; n=9)*
		Settlements total area (km^2)	60.48 [60.48–60.48] *(deterministic; n=9)*
		Mining total area (km^2)	4.62 [3.21–8.87] *(stochastic; n=9)*
		Mining interval new area (km^2)	2.27 [0.02–4.44] *(stochastic; n=9)*
		Oil & gas seismic line length (km)	10,749.95 [10,723.47–10,766.15] *(near-deterministic; n=9)*
UA sitesel (stochastic configuration uncertainty)	SiteSelectionAsDistributing varies across 4 configurations; held constant: totalDisturbanceRate=1.5, clusterDistance=1000, useClusterMethod=TRUE	Mining total area (km^2)	8.872 [8.872–8.872] *(deterministic; n=10 per design)*
		Forestry cutblocks total area (km^2)	246.129 [246.129–246.129] *(deterministic; n=10 per design)*
		Oil & gas total area (km^2)	6.313 [6.313–6.313] *(deterministic; n=10 per design)*
	siteSelectionAsDistributing = seismicLines	Seismic line length (km)	12,895.88 [12,832.13–12,954.70] *(n=10)*
	siteSelectionAsDistributing = oilGas and oilGas;cutblocks;mining	Seismic line length (km)	11,942.01 [11,918.15–12,020.91] *(n=10)*
	siteSelectionAsDistributing = oilGas;cutblocks;mining;seismicLines	Seismic line length (km)	12,916.67 [12,868.49–12,981.79] *(n=10)*
	(derived contrast)	Seismic regime gap (km) *(configs including seismicLines vs excluding)*	≈960 km *(dominates within-design stochastic spread)*
UA random (parametric/design uncertainty	Designs vary: totalDisturbanceRate 0.662–1.970, clusterDistance 501.7–1984.6; constant in this batch: useClusterMethod=TRUE, siteSelectionAsDistributing=seismicLines	Seismic line length (km)	8,663.31–11,839.02 *(range across design medians; typical within-design p05–p95 width ≈19 km)*

Two practical lessons emerge. First, several sectors are effectively deterministic at the aggregate level under fixed settings (e.g., settlements and cutblocks), meaning that additional replicates do not materially change their totals; in contrast, mining expresses substantial seed-driven variability under the default regime, and this stochastic spread is a meaningful part of the benchmark evidence (Table 1). Second, uncertainty is often regime-dependent rather than uniformly "noisy": seismic line outcomes can shift by configuration in a way that dominates within-configuration stochastic spread (Table 1). Finally, the parametric ensemble reported here is explicitly restricted to `useClusterMethod` = TRUE ("UA parametric-TRUE"), separating within-regime stochastic uncertainty from between-design variability and avoiding conflating a structural regime switch with ordinary parameter perturbations.

3.3 Validation against observations

In this benchmark I validate against BEAD observations over the NWT study area (100 m resolution) using two complementary targets: a standard mapped footprint and a ecological influence-zone footprint defined by a 500 m buffer. For each target I run replicated simulations and compute a set of metrics chosen explicitly to avoid common benchmarking pitfalls—most importantly, the tendency of overall map accuracy to appear high simply because "no change" dominates most pixels.

For the observation-based validation runs, the generator was exercised largely with module defaults; the primary ecological variant was the zone-of- influence interpretation, operationalized via buffering disturbances (500 m) using the `disturbanceRateRelatesToBufferedArea` = TRUE setting. An example validation configuration indicates the following notable settings (illustrative, not exhaustive): run interval of 5 years; rates derived from a fixed historical difference window (2010–2015); the clustering regime enabled with an explicit clustering distance; and a distributing allocation for seismic lines rather than exhaustively allocating to "best" sites.

The validation uses two scenarios with different benchmarking roles. First, an in-sample verification benchmark compares simulated 2010–2015 change to observed BEAD 2010→2015 change. This scenario is expected to be the easier test because the rate regime is derived from the same interval; it primarily checks that the pipeline, data crosswalk, and disturbance mechanisms can reproduce the calibration-era pattern. Second, an out-of-sample hold-out benchmark compares simulated change after the 2015 baseline to observed BEAD 2010→2020 change while keeping the 2010–2015 rate regime fixed. This scenario is the stronger corroboration test because it probes whether the calibrated regime generalizes to a later period; when it fails, the failure can be interpreted mechanistically as nonstationarity (rates or drivers changed) or as insufficient spatial allocation rules, rather than as a generic “bad model” conclusion.

Because disturbance change is sparse relative to the landscape, I report overall agreement for completeness but do not treat it as a headline skill metric. Instead, I emphasize three diagnostics that are more informative for ecologists benchmarking stochastic landscape generators. First, I compute change prevalence (the fraction of pixels that change) and binary footprint skill (precision/recall/F1 on change vs no-change). Second, I decompose disagreement using Quantity Disagreement (QD) and Allocation Disagreement (AD), and I report these both on the full map and on the change mask (restricted to pixels that changed in either map) to avoid background dominance. This QD/AD split is particularly useful as a “debugging lens”: high QD points toward a quantity or class-portfolio mismatch (e.g., rate regime or class mapping), whereas high AD points toward misallocation (e.g., missing spatial drivers or constraints). Third, because pixel-accurate overlap is often unrealistic for stochastic generators, I add scale-aware corroboration by aggregating disturbed fraction to 1 km, 5 km, and 10 km grids and reporting Spearman correlation, and I add a linear-distance corroboration (median and 90th percentile distances from observed linear-change pixels to the nearest simulated linear change).

In the standard footprint verification (2010–2015) benchmark ($n = 9$ replicates available), overall agreement is high (OA = 0.989), but this reflects the dominance of “no change” rather than strong predictive overlap. The more informative checks show that the model underproduces new disturbance: observed change prevalence is 1.12% of pixels while simulated prevalence is 0.55%, corresponding to an observed unique new-disturbance area of 6413 km^2 versus 3152 km^2 simulated (Fig. 2). This quantity shortfall is reflected in the disagreement decomposition: on the change mask the total disagreement is high ($D\_change = 0.807$) with $QD\_change = 0.471$ larger than $AD\_change = 0.335$. Binary change-detection skill is therefore recall-limited: predicted change pixels are frequently correct (precision = 0.537) but most observed change is missed (recall = 0.264), yielding F1 = 0.354. A useful benchmarking insight emerges when examining scale: although pixel overlap is modest, broad spatial gradients are reproduced more consistently at coarser resolution, with grid-based Spearman correlation increasing from ~0.43 (1 km) to ~0.58 (10 km) (Fig. 3). Linear-distance corroboration indicates substantial displacement for linear disturbances (median

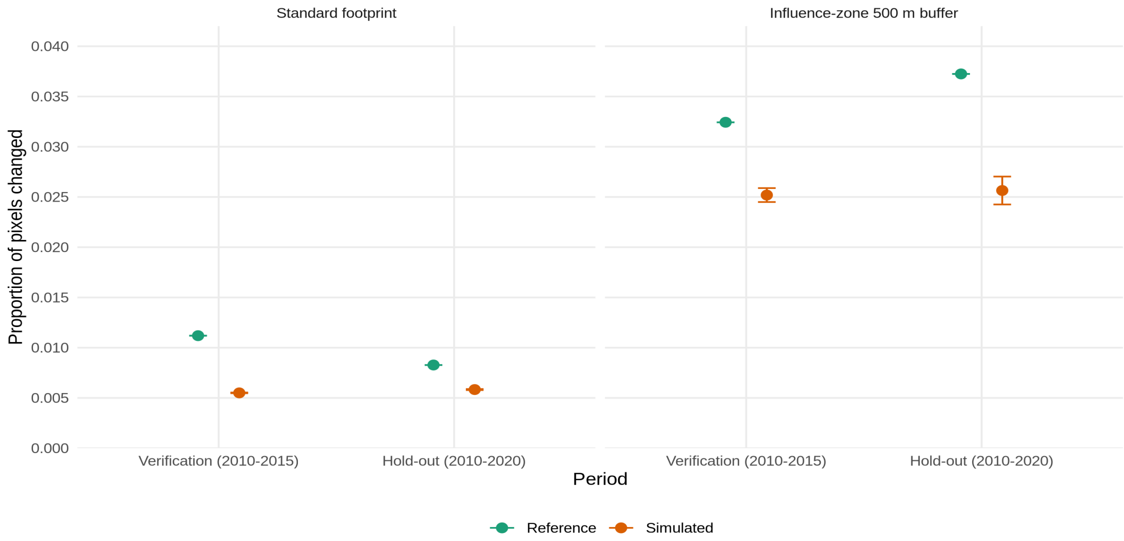

Figure 2: Change prevalence (new disturbance) in benchmark vs simulated disturbance maps for the verification period (2010–2015) and hold-out period (2010–2020) under two footprint interpretations (Standard footprint; influence-zone 500 m buffer). The y-axis is the proportion of pixels that transitioned to disturbed between the 2010 baseline and the end year. Reference points are computed directly from the benchmark maps; simulated points are the mean across stochastic replicates with error bars showing ±1 SD (n = 9–10 replicates, depending on scenario).

distance ~2.17 km), which is consistent with the well-known tendency for low allocation agreement in stochastic landscape generators.

In the standard footprint hold-out benchmark (2010–2020) (n = 10 replicates), the diagnosis shifts in an instructive way. Overall agreement remains high (OA = 0.988), but on the change mask disagreement is much higher (D_change = 0.934), and the decomposition becomes allocation-dominated (AD_change = 0.624 versus QD_change = 0.310). In other words, in hold-out the primary limitation is increasingly "where" rather than only "how much." The binary skill metrics also weaken (precision = 0.200, recall = 0.141, F1 = 0.166), while the prevalence gap persists (reference 0.83%, simulated 0.58%) (Fig. 2) and the unique disturbed area remains underpredicted (observed 4739 km^2, simulated 3341 km^2). Coarse-scale corroboration remains moderate but declines relative to verification (10 km ρ ~0.51) (Fig. 3), and linear displacement increases (median ~3.44 km). From a benchmarking perspective, this pattern is informative: the calibration-era rate regime can generate disturbance of roughly the right order of magnitude, but its spatial allocation and/or disturbance-type composition does generalize reasonably into 2010–2020, consistent with changing disturbance drivers or missing spatial covariates that become more important in the later period.

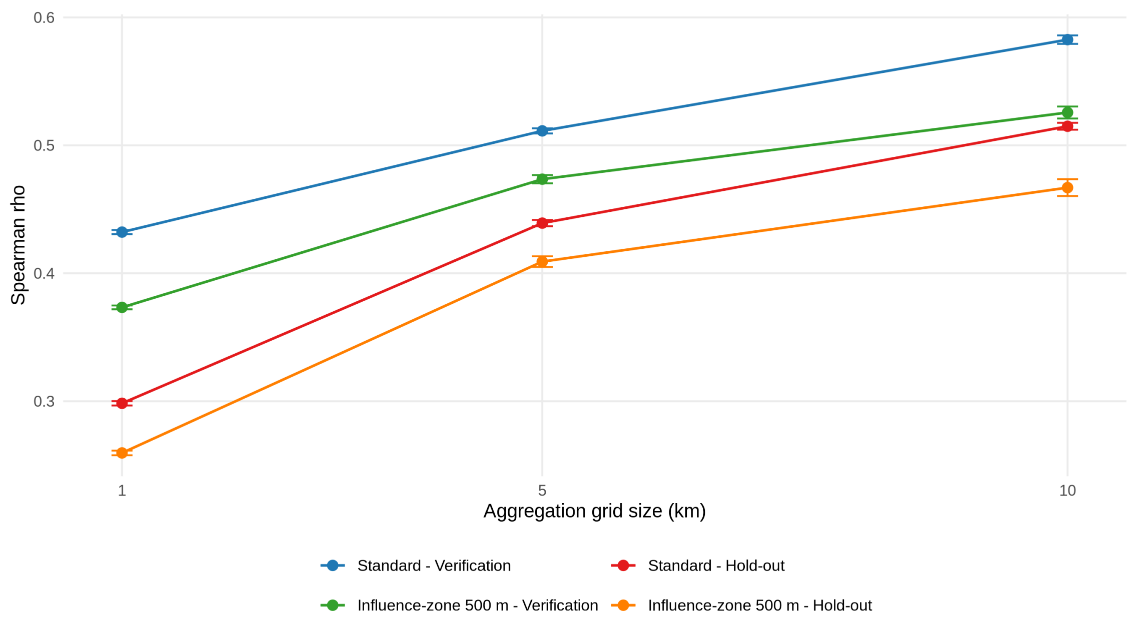

Figure 3: Scale-dependent corroboration of spatial disturbance patterns using Spearman rank correlation (ρ) between benchmark and simulated disturbed-fraction grids. Disturbed fraction is aggregated to 1 km, 5 km, and 10 km grid cells (x-axis), and ρ is computed per replicate for verification (2010–2015) and hold-out (2010–2020) under Standard and influence-zone 500 m footprint assumptions. Points show the mean across replicates with error bars ±1 SD (n = 9–10)..

Class-wise quantity diagnostics (unique footprint; crosswalk classes) provide a consistent mechanistic "signature" across both scenarios and help turn aggregate disagreement into actionable model-development insight. In verification, Seismic disturbance is strongly underpredicted (observed 5306 km^2 vs simulated 1616 km^2, ratio 0.30×), while Road is overpredicted (observed 840 km^2 vs simulated 1157 km^2, ratio 1.38×). In hold-out, the same imbalance persists and road overprediction intensifies: Seismic remains underpredicted (observed 4057 km^2 vs simulated 1771 km^2, ratio 0.44×), while Road is overpredicted by 2.47× (observed 471 km^2 vs simulated 1164 km^2). Several smaller classes (e.g., settlement, mine and cutblocks) are also inflated in relative terms. This recurrent portfolio shift indicates that disagreement is not only a matter of stochastic noise; it points to structural differences between how the generator expresses linear disturbance types and how those types appear in the observational product, which likely propagates into both quantity disagreement (too little seismic footprint) and allocation disagreement (misplaced linear features).

I then repeated the benchmark under a 'zone of influence' scenario by buffering disturbances to 500 m, as described in the ODD. This buffering was applied consistently at the evaluation stage to assess whether the model's unbuffered outputs successfully reproduce broad spatial impact patterns beyond the immediate feature boundaries. This benchmark target is intentionally different: it asks whether the model reproduces influence zones relevant to conservation (i.e., boreal woodland caribou) with a focus on intactness metrics, not whether it reproduces mapped disturbance widths. As expected, buffering increases change prevalence and reduces the dominance of "no change," making disagreement more stringent. In the 500 m influence-zone verification (2010–2015), reference prevalence is 3.24% versus 2.52% simulated, corresponding to observed buffered disturbed area 18,579 km² versus 14,430 km² simulated (≈ −22%). Binary footprint skill is low (F1 = 0.169), and disagreement on the change mask is overwhelmingly allocation-driven (D_change = 0.965, AD_change = 0.672). Hold-out influence-zone benchmarking further degrades (reference prevalence 3.72% vs simulated 2.56%, buffered disturbed area 21,318 km² vs 14,689 km², ≈ −31%; F1 = 0.133, D_change = 0.984). Coarse-scale corroboration remains the most consistently positive signal (10 km ρ ~0.53 in verification and ~0.47 in hold-out), but linear displacement grows (median ~4.48 km and ~5.47 km, respectively). At the 10 km aggregation used, the map comparison (Fig. 4) reveals where this mismatch concentrates spatially, with coherent areas of under- and over-predicted disturbed fraction despite moderate rank-correlation at coarse grain. Finally, the non additive per-class buffered-footprint diagnostics (each class treated as its own influence zone rather than an additive total) reinforce the same seismic–road substitution seen under the standard footprint: buffered seismic influence is underrepresented (≈ 0.40–0.46× observed), while buffered road influence is strongly overrepresented (≈ 3.4–3.6× observed).

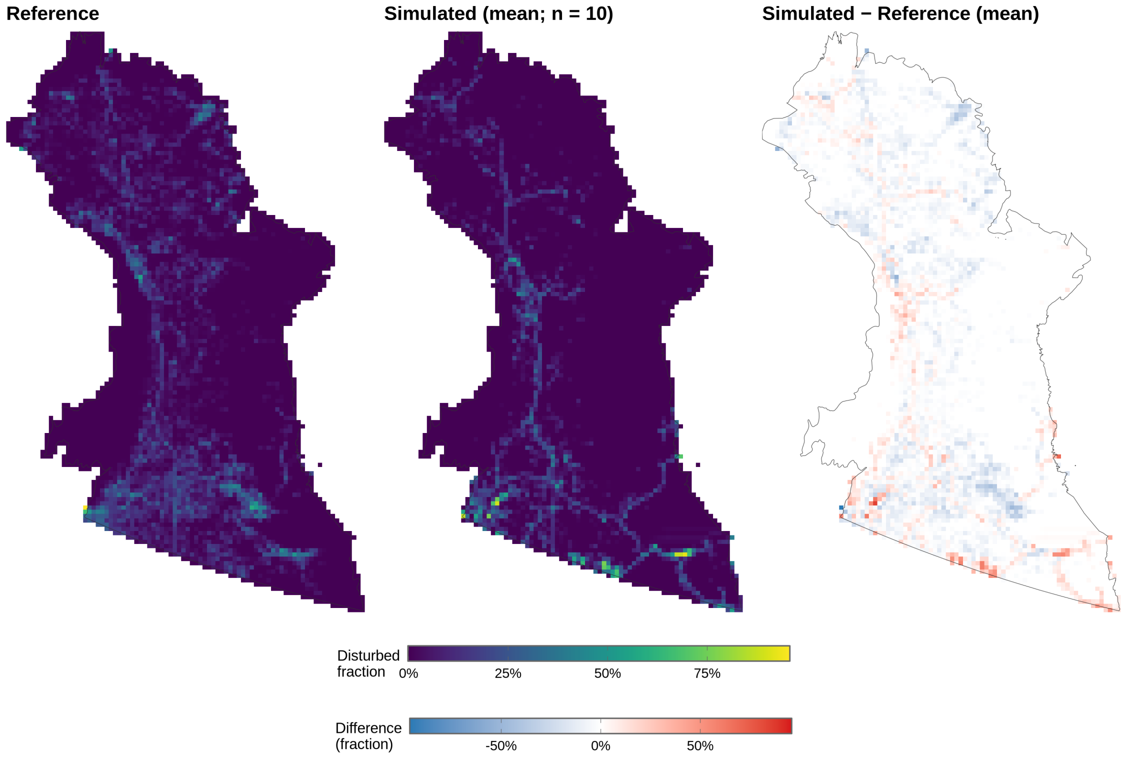

Figure 4: Grid-aggregated anthropogenic disturbance change for the hold-out period (2010–2020) under a 500 m influence-zone footprint. Left: reference disturbed fraction per 10 × 10 km cell from the benchmark maps (BEAD); center: simulated disturbed fraction (ensemble mean; n = 10) processed with the same buffering and aggregation; right: bias map (simulated − reference), where red indicates overprediction and blue underprediction of disturbed fraction.

Taken together, this benchmarking exercise yields three practical conclusions that guide both interpretation and future model refinement. First, headline accuracy must be change-focused in sparse-change landscapes; change-mask disagreement and binary footprint skill provide a more honest assessment than overall agreement. Second, the model exhibits a persistent disturbance-portfolio mismatch (notably seismic underprediction and road overprediction) that is consistent across scenarios and is amplified under a 500 m influence-zone benchmark, indicating a structural target for refinement. Third, while pixel-level corroboration is as expected weak—especially in hold-out—there is moderate corroboration of broad spatial gradients at 5–10 km (Fig. 4), suggesting that in its current form the generator is suited as a scenario generator for coarse-scale disturbance patterns.

3.4 Traceability of benchmarking evidence

To support auditability of the benchmarking results in Sections 3.1–3.2, Appendix B provides a traceability matrix linking ODD-derived model claims (Appendix A) to explicit acceptance rules and concrete evidence artifacts (e.g., unit-test outputs, configuration validation, and parsed summary metrics). In the current snapshot, all unit tests pass (0 failures; 4 skipped), with overall test coverage ≥70%, indicating strong verification coverage of parameterisation and scheduling logic. The matrix also shows stable execution of the core benchmarking pipelines: all experiment suites completed without run failures (system 4/4; sensitivity 240/240; uncertainty 82/82; 100% completion). Evidence types differ in how routinely they are re-checkable. Lightweight verification artifacts (unit tests and configuration checks) are CI-suitable and can be rerun frequently, whereas computationally expensive sensitivity and uncertainty ensembles are recorded as versioned snapshot evidence and interpreted alongside documented design caveats. Overall, the traceability matrix provides a structured mapping from benchmark conclusions to reproducible artifacts and acceptance rules, enabling consistent comparison across future reruns and releases.

4 Discussion

Across the evaludation sequence, the *anthropogenicDisturbance* set of modules proved technically reproducible (pinned inputs, scripted builds, continuous integration) and logically consistent with its description and stated requirements, while sensitivity scouting and uncertainty ensembles clarified which configuration choices materially affect disturbance regimes. Validation against observed disturbance baselines showed that agreement is strongest for quantity- and rate-based benchmarks and for broad spatial structures, whereas pixel-level overlap is limited in sparse-change landscapes and compositional mismatches persist between disturbance types (notably roads versus seismic lines). The following sections interpret these results as benchmarking evidence: first clarifying why scenario generators should be assessed on process fidelity rather than pixel-perfect prediction, then using the quantity/allocation decomposition to identify failure modes, and finally translating the main mismatches into actionable priorities for model refinement and for downstream ecological applications.

4.1 Benchmarking a scenario generator, not a pixel-accurate predictor

A central outcome of this work is a clearer statement of what kind of scientific claim a stochastic disturbance generator can and cannot support. The *anthropogenicDisturbance* set of modules is designed to generate plausible disturbance scenarios constrained by empirical rates and sectoral patterns, not to reconstruct the exact footprint of historical projects. Under this purpose, credibility hinges on whether the implemented processes and constraints produce realistic ensembles of futures, and on whether the evidence linking assumptions to outputs is transparent and reproducible (Augusiak et al., 2014; Schmolke et al., 2010).

Accordingly, validation results are interpreted through an evaludation lens: the most important question is not whether a single map matches pixel-by-pixel, but whether benchmarking evidence is consistent across data evaluation, conceptual expectations, verification tests, sensitivity/uncertainty analysis, and corroboration. This multi-evidence framing also makes it explicit where the current implementation is robust, and where it is likely to benefit from targeted refinement.

4.2 The accuracy paradox in sparse-change landscapes

A key lesson from validating sparse land-change processes is that apparent “low accuracy” at the pixel scale is often an artefact of the benchmark, not a failure of the model. This paradox is evident in our own diagnostics: map-wide agreement can appear reassuring, while the change-focused evaluation within the disturbance footprint reveals substantial mismatch that is obscured when ‘no-change’ dominates the map (Fig. 2). When change occupies a small fraction of the landscape, agreement is dominated by persistence and chance alignment, so pixel-overlap metrics can be both unstable and misleading as indicators of decision relevance (Pontius & Millones, 2011; Pontius et al., 2008). For stochastic landscape generators, demanding pixel-perfect prediction is therefore an impossible standard: the appropriate question is whether the model reproduces the right patterns at the scales that matter for inference and management (Levin, 1992; Grimm et al., 2005; Grimm & Railsback, 2012).

The interpretive move is to treat low fine-grain overlap as expected and to prioritize scale-aware, process-aligned benchmarks. In practice, this means emphasizing quantity and rate constraints, composition (the disturbance “portfolio”), and spatial structure summaries (e.g., clustering, edge density, buffered footprints), and using multiple criteria so that different failure modes become diagnostically separable (quantity versus allocation; amount versus pattern).

the grid-aggregated comparison in Figure 4 demonstrates that the model successfully captures the broader disturbance regimes of the NWT. The visual alignment between the reference (left) and simulated (center) maps confirms that the model is 'right for the right reasons' at a landscape level, correctly identifying the primary corridors of development even when the exact coordinate of a disturbance event is displaced.

4.3 External context: what counts as “good” in land-change model validation

Placing the results in the broader land-change modelling literature helps avoid an overly internal reading of performance. Spatially explicit land-use/land-cover (LULC) and land-change models—particularly stochastic or rule-based allocators—are widely documented to struggle with cell-by-cell replication of observed change, even when they are useful for scenario analysis and pattern reproduction at aggregated scales (Verburg et al., 2002; Verburg et al., 2004). Comparative studies emphasize that, at native resolution, correctly predicted change is often small relative to the amount of disagreement, and that validation must therefore be interpreted with respect to the model’s purpose and the spatial/temporal grain of the decision question (Pontius et al., 2008; Pontius & Petrova, 2010).

This context supports two interpretation principles that guide the remainder of the Discussion:

1. **Moderate, scale-aware skill can be meaningful (Fig. 3).** The generator reproduces broad spatial tendencies (e.g., where disturbance pressure is higher) at a coarse grid (Figure 4). This is a strong outcome for an ecological model that depends on exposure surfaces and relative risk patterns rather than exact footprints. This is consistent with long-standing recommendations that land-change validation should be explicitly scale-aware and pattern-oriented when the modelling objective is scenario exploration (Verburg et al., 2004; Pontius & Petrova, 2010).

2. **Low overlap does not preclude ecological usefulness.** For sparse-change processes, low overlap metrics often reflect the combinatorial difficulty of predicting exact locations of rare events, compounded by observation uncertainty. A more informative question is whether the model reproduces the amount of change, its clustering/dispersion, and the portfolio of disturbance types relevant to ecological endpoints (Pontius & Millones, 2011; Pontius et al., 2008).

These principles also clarify why benchmarking should not collapse into a single headline score: different metrics illuminate different failure modes (quantity vs allocation; amount vs pattern).

4.4 Portfolio mismatch: when the regime is plausible but the sector mix is not

One of the clearest diagnostic findings is that the generator can reproduce a broadly plausible overall disturbance regime while still misrepresenting the sector mix. Across the benchmarking period, quantity-based results indicate that some disturbance classes are systematically over- or under-produced relative to observations, which implies that correct totals at the aggregate level can mask compensating errors among disturbance types.

In the current model structure, roads are not allocated as an independent disturbance class. Instead, road-length targets are implicitly coupled to the generation of other polygon disturbances (e.g., access associated with resource extraction), so a bias in one sector can propagate into an overproduction of roads unless additional constraints are introduced (e.g., avoidance rules, project-type-specific road densities, or explicit road network generation).

Seismic lines, by contrast, are generated using a clustering-based allocation approach. This design can reproduce local concentrations but is sensitive to configuration choices (e.g., clustering method, distance thresholds, and site-selection rules), and it can underproduce the dispersed, low-density footprint observed in some periods.

Two implementation-level mechanisms remain especially relevant as general explanations for portfolio mismatch:

- Distributed versus exhaustive allocation. The module supports both distributed allocation (spreading disturbance across candidate sites) and more exhaustive placement modes. When distributed allocation is combined with strict clustering, some classes can become too spatially concentrated, increasing allocation disagreement and reducing coverage of dispersed disturbance.
- Clustering regimes and distance thresholds. The clustering configuration strongly controls the fragmentation and connectivity of linear disturbances. Small changes in distance thresholds can shift the generator between dispersed and clustered regimes, affecting both total linear length and where disturbance accumulates.

One benchmarking lesson is to treat compositional mismatch as a distinct failure mode: it directly indicates which mechanisms require refinement, and it helps prioritize follow-up tests (e.g., targeted experiments on site selection, clustering, and coupling assumptions) before adjusting parameters elsewhere.

This portfolio mismatch matters biologically because different linear-feature types do not have equivalent ecological effects. For example, roads can function as movement barriers for caribou at moderate traffic levels, whereas seismic lines may be more permeable but can increase predator access and contribute to functional habitat loss over large areas (Dyer et al., 2002; Environment Canada, 2011). Consequently, over-predicting roads while under-predicting seismic lines can bias downstream habitat-impact indicators that depend on the density, width, and buffering of linear features, even if total disturbed area appears reasonable.

4.5 Disturbance influence-zones: why the buffer scenario matters

Because the intended downstream use includes forecasting habitat change for boreal caribou, validating raw footprint polygons alone is not sufficient: the ecologically relevant exposure signal often comes from zones of influence around anthropogenic features. The use of a 500 m buffer around anthropogenic footprint is not arbitrary—it is closely aligned with national-scale critical habitat assessment practice for boreal caribou, where buffered disturbance is used as a standardized approximation of functional habitat loss at broad scales (Environment Canada, 2011; Environment and Climate Change Canada, 2012).

At the same time, the ecological literature makes clear that behavioural responses and predator-mediated effects can extend beyond narrow buffers and vary by disturbance type and context. Empirical studies document avoidance and movement effects relative to industrial features (including linear corridors) at distances that can exceed several hundred metres (Dyer et al., 2001), and restoration work emphasizes that the functional influence of linear features depends on vegetation recovery and predator movement dynamics, not only on the physical footprint (Dickie et al., 2017). A practical interpretation is that the 500 m buffer is best viewed as a policy-relevant, conservative standardization for comparing scenarios, rather than a definitive behavioural threshold.

This framing also clarifies the modelling contribution: the disturbance generator is most defensible as a scenario generator that provides consistent, reproducible disturbance layers from which buffered exposure surfaces can be derived and compared across climate and management futures, rather than as a pixel-accurate predictor of the exact location of new disturbances.

4.6 Regime-dependent variability and near-determinism

The uncertainty analysis reinforces that the generator's variability is regime-dependent: some disturbance components behave nearly deterministically under fixed inputs, while others show broader stochastic spread (Table 1). This aligns with good practice in environmental modelling, where uncertainty assessment is expected to distinguish stochastic variability from structural and parametric uncertainty, and to focus computational effort where it is epistemically justified (Pianosi et al., 2016; Saltelli et al., 2007). For users, the key implication is operational: ensemble thinking is most valuable where the generator is genuinely stochastic or where configuration choices create regime switches (e.g., clustering methods) (Fig. 1).

A second, more diagnostic implication emerges from the near-deterministic behaviour observed in several polygonal disturbance classes: where a class is configured to be stochastic (e.g., stochastic placement, stochastic growth, or stochastic selection among candidates) yet produces effectively identical outcomes across seeds, it is important to identify why the stochasticity is not propagating to the output. In benchmarking terms, this is not merely a curiosity; it determines whether an ensemble is providing meaningful uncertainty information or simply repeating the same mechanistic outcome. Near-determinism can occur for benign reasons—for example, if polygonal outcomes are dominated by hard constraints or saturating masks that leave little freedom for random placement—but it can also indicate that the intended stochastic pathway is being short-circuited by implementation details.

Accordingly, a useful extension of the uncertainty analysis is to treat these polygonal classes as a targeted follow-up test case: confirm that stochastic switches are enabled as intended; instrument the workflow to record (per class, per interval) the number of candidates, the number of stochastic draws actually executed, and whether ties are resolved deterministically; and verify that different seeds lead to different intermediate decisions even if aggregate area totals remain stable. This is precisely the type of "cause-of-(non)variability" check that strengthens benchmarking under compute constraints: it prevents over-interpreting stable ensemble summaries as evidence of low uncertainty when the underlying reason is simply that randomness is not operational in that pathway, and it helps focus model refinement on the few decision points where stochasticity should matter most (Pianosi et al., 2016; Saltelli et al., 2007).

4.7 Why verification and traceability deserve space in ecological papers

Calls for reproducibility and transparency are now common in ecological modelling, but the adoption of good modelling practices remains uneven because it is shaped by academic incentives and publication norms, not only by the availability of protocols—hence the case for explicitly making verification and traceability visible in ecological papers (e.g., brief verification statements, traceability summaries, and links to test suites and archived run logs) rather than treating them as optional appendices (Micheletti et al., 2024). A recurring lesson in this project is that verification work is rarely "shiny," but it consumes a large share of real modelling effort—and it is often the difference between a model that is merely runnable and one that is scientifically defensible. Complex spatial simulations are particularly vulnerable to subtle failures (data issues, topology errors, brittle assumptions) that may only be revealed when diverse parameter combinations and datasets are exercised. In this thesis, several issues surfaced late precisely because the benchmarking suite expanded the diversity of scenarios and exposed under-tested branches of logic.

This experience aligns strongly with predictive ecology frameworks that robust workflows (testing, continuous integration, reusable pipelines) are not software engineering luxuries but mechanisms that improve scientific reliability and speed up learning (McIntire et al., 2022). The practical lesson for ecologists is that verification artifacts—unit tests, integration tests, traceability tables, and reproducible run logs—should be treated as part of the scientific record that supports model claims, not as ancillary implementation details.

4.8 Limitations and threats to validity

Several limitations help interpret the results and define the most defensible scope of inference:

- **Spatial inhomogeneity vs global rates.** The study area is large and heterogeneous, with strong gradients in existing infrastructure density (notably higher in the south and lower in the north). Deriving rates as a single study-area summary can therefore produce systematic regional bias: a model that matches the mean can still misrepresent local regimes. A clear next step is stratified rate derivation (e.g., by ecozone, administrative units, or caribou monitoring areas) or the use of spatially varying rate surfaces, which is common in land-change modelling when drivers and constraints vary across space (Verburg et al., 2002; Verburg et al., 2004).
- **Forecast horizon.** The empirical benchmarking horizon was limited (e.g., to 2031) by compute and data constraints. However, many ecological forecasting applications run several decades to a century, particularly when coupled to vegetation and fire dynamics. SpaDES-based forecasting in the Northwest Territories has been applied over multi-decadal horizons for biodiversity and caribou demography (Micheletti et al., 2021; Stewart et al., 2023). Extending

benchmarking beyond the near term is therefore a priority, but it likely requires either more compute or more selective benchmarking designs.

- **Temporal dynamics of disturbance recovery.** Some disturbance types (notably seismic lines and forestry cutblocks) are temporally dynamic in their ecological effect: vegetation recovery can take decades and may be incomplete in some settings. Evidence from the boreal plains indicates wide seismic lines can persist with low woody recovery even after multiple decades (Lee & Boutin, 2006), and functional recovery for caribou depends on whether vegetation structure reduces predator travel efficiency (Dickie et al., 2017). Treating these disturbances as permanent is therefore conservative for exposure accounting but may overstate long-term impact unless a fading or recovery mechanism is implemented (e.g., decay functions or state transitions based on time-since-disturbance and site conditions).
- **Saturation and non-stationarity in high-density regions.** In parts of the southern study area, very high existing linear feature density can create saturation behaviour: even if derived rates are high, the allocator may be constrained by available space and rules that prevent unrealistic overlap. In addition, rates can be non-stationary across observation windows (e.g., higher rates during one period and lower during another), which can propagate directly into forecast mismatch if a single historical window is used for parameterization.
- **Equifinality under compute constraints.** With limited ability to run large calibration ensembles, multiple parameter combinations may produce similarly "moderate" agreement at aggregated scales. This is a classic equifinality challenge in environmental models and should temper claims about uniqueness of any one parameterization (Beven, 2006).

4.9 Benchmark-guided priorities for improving the disturbance generator

Benchmarking converts "model–data mismatch" into a prioritized development agenda. Based on the evidence assembled here, several improvements would likely yield the largest gains in ecological usefulness:

1. Stratify or regionalize rates and constraints to respect strong north–south heterogeneity and sector-specific development regimes.

2. Target seismic line configuration experiments (cluster regimes, distance thresholds, distributed site selection, and related parameters) to address consistent under-representation while keeping polygon totals plausible.
3. Refine the road-generation mechanism by testing alternative connectivity heuristics and constraint layers so that secondary road creation does not inflate accessibility unrealistically in remote contexts.
4. Implement disturbance recovery / fading mechanisms for temporally dynamic disturbances (e.g., seismic lines, cutblocks), aligned with caribou functional recovery concepts and time-since-disturbance evidence.
5. Extend benchmarking horizons and ensemble designs toward multi-decadal forecasts, leveraging the "scouting" results from sensitivity analysis to focus compute on high-leverage regimes.
6. Implement diagnostic checks to verify stochastic pathways in polygonal disturbance classes, ensuring intended variability and identifying implementation shortcuts.

The overarching conclusion is that the disturbance generator is best used—and best improved—as a scenario-consistent disturbance and exposure generator: it supports ecological forecasting by producing reproducible disturbance layers whose aggregated patterns and buffered influence zones can be meaningfully compared across futures, even when exact footprints remain inherently difficult to reproduce.

5 Conclusion

I set out to embed a complex ecological simulation module within a transparent benchmarking workflow. By addressing three research questions spanning verification, validation, and uncertainty, the analysis yields specific insights into both the *anthropogenicDisturbance* set of modules and the practice of ecological modelling.

Regarding the benchmarking workflow (RQ1), this study demonstrated that credibility in complex simulation models stems from traceable evidence rather than asserted authority. By coupling an ODD-based conceptual description with an automated test suite and a traceability matrix, I established an "evidence ledger" that links high-level model claims to concrete implementation checks. This approach proved that rigorous software verification—often neglected in ecological modelling—is feasible and essential for distinguishing between data limitations and implementation errors.

Regarding validation and spatial patterns (RQ2), the results highlight the "accuracy paradox" inherent in sparse-change landscapes: high overall agreement metrics masked significant limitations in predicting new disturbance locations. While the model struggled with pixel-perfect precision (F1 scores < 0.36), it corroborated observed spatial gradients at coarser scales (5–10 km), suggesting it is effective for generating regional exposure surfaces. However, the decomposition of disagreement revealed a structural "portfolio mismatch," specifically a systematic under-representation of seismic lines and an over-representation of roads. This indicates that the current allocation algorithms prioritize connectivity at the expense of the specific clustering regimes required to replicate exploration-based linear features.

Regarding uncertainty and robustness (RQ3), the analysis showed that model variability is regime-dependent rather than uniform. Some sectors, such as settlements and cutblocks, behaved near-deterministically under fixed parameters, while mining and seismic lines exhibited significant stochastic spread. Sensitivity screening further identified that structural switches (such as the clustering method) act as dominant controls on landscape configuration, overriding minor parameter tuning. This implies that future improvements should focus on structural refinements to the linear feature generator rather than exhaustive calibration of scalar rates.

The *anthropogenicDisturbance* generator is ultimately best understood not as a pixel-accurate predictor of the future, but as a defensible scenario generator capable of producing spatially consistent disturbance patterns for cumulative effects assessment.

By making the model's limitations explicit—specifically the trade-offs between connectivity and clustering—the present work supports its responsible use in decision contexts, such as caribou habitat planning, where relative risk and exposure surfaces are more critical than exact footprint location. More broadly, this work illustrates that "validity" in predictive ecology is not a binary state, but a documented argument supported by diverse lines of evidence. As ecological forecasts increasingly inform policy, the rigorous, automated, and transparent benchmarking practices detailed here offer a necessary path toward models that are not only powerful, but trustworthy.

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Declaration on the use of AI tools

In preparing this thesis, I made limited and transparent use of generative AI tools as a support aid in literature review, code review and text draft generation. I take full responsibility for the thesis in its entirety. Tools in use were OpenAIs GPT 5, Mistrals LeChat and Deepseek 3.

All AI-assisted outputs were treated as non-authoritative: I independently verified claims, validated code behaviour through testing, and ensured that citations refer to original sources that I read and assessed.